

Hypertension: Pathophysiology, Diagnosis, and Management

Hypertension, commonly called high blood pressure, is a chronic condition in which the force of blood against the arterial walls is persistently elevated. It is one of the leading modifiable risk factors for cardiovascular disease worldwide and is often called the “silent killer” because it frequently produces no symptoms until target-organ damage has occurred.

Classification of Blood Pressure

Blood pressure is recorded as two numbers: systolic pressure (during heart contraction) over diastolic pressure (during relaxation), measured in millimeters of mercury (mmHg). Under the 2017 ACC/AHA guideline, blood pressure is categorized as follows. Normal is below 120/80 mmHg. Elevated is a systolic of 120 to 129 with a diastolic below 80. Stage 1 hypertension is 130 to 139 systolic or 80 to 89 diastolic. Stage 2 hypertension is at or above 140 systolic or 90 diastolic. A hypertensive crisis is a reading above 180/120 mmHg and requires prompt medical attention.

Pathophysiology

Arterial pressure is the product of cardiac output and systemic vascular resistance. Any factor that increases either of these raises blood pressure. A central regulator is the renin–angiotensin–aldosterone system (RAAS). The kidneys release renin, which converts angiotensinogen to angiotensin I. Angiotensin-converting enzyme (ACE) then converts angiotensin I to angiotensin II, a potent vasoconstrictor. Angiotensin II also stimulates the release of aldosterone, which promotes sodium and water retention, increasing blood volume. The sympathetic nervous system contributes by increasing heart rate and vasoconstriction.

Primary vs Secondary Hypertension

Roughly 90 to 95 percent of cases are primary (also called essential) hypertension, which has no single identifiable cause and develops gradually over years. The remaining 5 to 10 percent are secondary hypertension, caused by an underlying condition such as chronic kidney disease, renal artery stenosis, primary aldosteronism, pheochromocytoma, or the use of certain medications. Identifying secondary causes matters because treating the root condition can cure the hypertension.

Risk Factors and Complications

Major risk factors include advancing age, obesity, a high-sodium diet, excessive alcohol intake, physical inactivity, smoking, and family history. Left untreated, sustained hypertension damages organs over time, leading to stroke, myocardial infarction, heart failure, chronic kidney disease, and hypertensive retinopathy.

Management

First-line management emphasizes lifestyle modification: the DASH diet (rich in fruits, vegetables, and low-fat dairy), reducing dietary sodium to under 2.3 grams per day, regular aerobic exercise, weight loss, and moderation of alcohol. When medication is required, common first-line drug classes are thiazide diuretics, ACE inhibitors, angiotensin II receptor blockers (ARBs), and calcium channel blockers. ACE inhibitors and ARBs are particularly favored in patients with diabetes or chronic kidney disease because they protect the kidneys. Therapy is individualized, and many patients require a combination of two or more agents to reach their target blood pressure.